PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: MediFlow

Version: 1.0

1. Project Overview

- **Problem Statement:** Healthcare access is often slowed by inefficient and manual triage processes. Patients struggle to accurately describe their symptoms, leading to misclassification of urgency and long waiting times—especially in overcrowded public healthcare systems such as Klinik Kesihatan and government hospitals in Kuala Lumpur. Additionally, existing appointment booking systems lack intelligent decision-making, forcing patients to self-assess urgency and choose doctors without medical guidance. This results in:
  - Delayed treatment for critical cases
  - Overloaded healthcare facilities
  - Poor patient experience due to unclear navigation

- **Target Domain:** AI-powered healthcare workflow automation, specifically focused on:
  - Outpatient triage and early symptom assessment
  - Intelligent appointment scheduling systems
  - Decision-support tools for patient self-navigation

  This solution targets early-stage healthcare interaction, where fast, accurate triage and guidance can significantly improve system efficiency and patient outcomes.

- **Proposed Solution Summary:**

  MediFlow is an AI-driven hospital triage and appointment automation system that transforms unstructured patient input into structured medical insights and actionable decisions.

  Using a stateful multi-agent architecture, MediFlow:
  - Interprets patient symptoms through natural language
  - Classifies urgency levels using a hybrid AI + rule-based triage system
  - Dynamically asks follow-up questions to resolve incomplete information

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- Automatically recommends appropriate doctors and schedules appointments

Instead of acting as a simple chatbot, MediFlow functions as an intelligent healthcare navigator, shifting the user's role from uncertain decision-maker to guided participant in their own care journey.

The system bridges the gap between patients and healthcare services by delivering:

- Faster triage decisions
- Reduced waiting time
- More accurate and structured medical case handling

2. Background & Business Objective

- **Background of the Problem:** Traditional healthcare systems rely heavily on manual triage and fragmented appointment workflows. Patients must describe their symptoms to front-desk staff or select departments themselves, often without medical knowledge. This creates a linear and inefficient process where:
  - Symptom interpretation depends on human judgment, leading to inconsistency
  - Critical cases may be delayed due to misclassification
  - Non-urgent patients overload emergency or specialist services
  - Appointment booking systems operate independently, without a medical context

  In high-demand public healthcare environments such as Klinik Kesihatan and hospitals in Kuala Lumpur, this results in long waiting times, overcrowding, and reduced quality of care.

  Additionally, existing digital health platforms lack intelligent interaction; they function as static booking tools rather than systems capable of understanding patient conditions and guiding decisions.

- **Importance of Solving This Issue:** Improving early-stage triage and patient navigation can significantly enhance healthcare efficiency and outcomes. By automating symptom understanding and urgency classification:
  - Critical cases can be identified and escalated immediately
  - Non-urgent cases can be redirected appropriately, reducing system overload
  - Waiting times can be minimized through better prioritization

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- Patients receive clearer guidance, improving overall experience

Most importantly, it reduces reliance on patients' self-assessment, which is often inaccurate, and replaces it with data-driven, structured decision-making. This enables faster, safer, and more scalable healthcare service delivery.

- **Strategic Fit / Impact:** MediFlow aligns with the growing shift toward AI-driven healthcare automation and decision support systems. By leveraging a multi-agent architecture with stateful interactions, it reflects modern AI system design principles where multiple specialized agents collaborate to simulate real-world workflows.

  The solution also fits within an AI-native development ecosystem, where:
  - AI models handle natural language understanding and triage reasoning
  - Backend systems manage state, workflows, and scheduling logic
  - Modular components allow easy deployment, scaling, and integration with healthcare infrastructure

  From an impact perspective, MediFlow:
  - Enhances operational efficiency in healthcare institutions
  - Reduces workload on frontline staff
  - Improves patient safety through early risk detection
  - Serves as a scalable foundation for future smart hospital systems

3. Product Purpose

3.1. Main Goal of the System:

To provide an AI-powered healthcare assistant that automates patient triage and appointment decision-making. MediFlow aims to transform unstructured patient input into accurate urgency classification, structured medical insights, and guided healthcare actions, enabling faster and safer access to medical services. Rather than acting as a simple booking tool, the system functions as an intelligent decision-support partner, helping users navigate healthcare services with clarity and confidence.

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3.2. Intended Users (Target Audience):

1. Patients (Primary Users)

- Individuals seeking medical guidance based on their symptoms
- Users who are unsure about the urgency of their condition
- Patients who want a fast and guided way to book medical appointments

MediFlow is designed to assist patients in:

- Understanding their symptoms through AI interpretation
- Receiving urgency classification (Emergency, Urgent, Non-Urgent)
- Getting appropriate medical recommendations and appointment scheduling

2. Healthcare System (Indirect Beneficiaries - *not direct users in current MVP***)**

- Clinics and hospitals that benefit from improved patient distribution
- Healthcare providers are experiencing reduced overload from non-urgent cases
- Emergency departments are receiving better-prioritized cases

Although MediFlow currently focuses on the patient-side interface, it indirectly supports healthcare institutions by:

- Reducing unnecessary hospital visits
- Improving patient routing
- Minimizing delays in critical cases

4. System Functionalities

4.1. Description: MediFlow operates as an AI-driven multi-agent healthcare workflow system that transforms unstructured patient input (text, voice, or form-based) into structured medical insights and actionable decisions.

Using a stateful, agent-based pipeline, the system performs:

- Natural language understanding of symptoms
- Real-time urgency classification (triage)
- Dynamic follow-up questioning for incomplete data
- Intelligent doctor recommendation and appointment scheduling

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Rather than functioning as a static chatbot, MediFlow acts as a decision-support engine, simulating real-world hospital triage workflows and guiding patients step-by-step toward appropriate medical care.

4.2. Key Functionalities:

- **AI Symptom Understanding:** Translates human-language symptom descriptions into structured medical data, including symptoms, duration, severity, and risk indicators.
- **Adaptive Conversational Triage:** Facilitates continuous interaction with the user by asking dynamic follow-up questions when information is incomplete, enabling more accurate and stateful medical assessment.
- **Stateful Case Tracking:** Maintains and updates patient information throughout the session, allowing the system to refine decisions progressively and generate a consistent, structured medical case.
- **Intelligent Appointment & Care Routing:** Automatically recommends appropriate actions based on urgency level, including emergency guidance, fast-track booking, or standard appointment scheduling with relevant doctors.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: MediFlow uses Z.AI GLM (ilmu-glm-5.1) as the core large language model.

Justification:

- Strong natural language understanding for medical symptom descriptions
- Capable of structured output generation (JSON-style responses)
- Efficient performance suitable for real-time interaction
- Supports multi-step reasoning required for triage and follow-up questioning

This makes it suitable for handling health-related conversational workflows with structured decision outputs.

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4.3.2. Prompting Strategy: The system adopts a multi-step agentic prompting approach, combining:

- Zero-shot prompting for initial symptom understanding
- Role-based prompts for each agent (triage, follow-up, scheduling)
- Sequential prompting where outputs from one agent become inputs for the next

Why this fits:

- Enables modular decision-making (each agent has a clear role)
- Improves accuracy by breaking complex tasks into smaller steps
- Supports stateful interactions (important for follow-up questioning)

4.3.3. Context & Input Handling: MediFlow handles unstructured and variable-length inputs through:

- Preprocessing Layer:
    - Cleans and standardizes text input
    - Extracts key entities (symptoms, duration, severity)
- Input Size Handling:
    - Accepts moderate-length user descriptions (typical conversational input)
    - For oversized input:
        - Important medical keywords are extracted
        - Irrelevant or repeated content is trimmed
- State Management:
    - Stores extracted data instead of raw long text
    - Ensures efficient context passing between agents

If input exceeds limits:

- System prioritizes critical medical information
- Non-essential data is truncated to maintain performance

4.3.4. Fallback & Failure Behavior: To ensure reliability and safety in healthcare-related decision-making, MediFlow implements multiple fallback and failure-handling mechanisms when the AI model produces off-topic, incomplete, or potentially hallucinated responses.

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1. Output Validation Mechanism: All AI responses are validated against expected formats (e.g. presence of triage level, structured fields). If the response is missing key information or appears inconsistent, it is flagged as unusable.

2. Automatic Retry Strategy: When an invalid or off-topic response is detected, the system automatically re-prompts the model with refined instructions to regenerate a more accurate and structured output.

3. Rule-Based Safety Override: Critical symptoms (e.g. chest pain, shortness of breath) trigger predefined high-risk rules. This ensures that even if the AI model underestimates severity, the system will still classify the case conservatively (e.g. as urgent or emergency).

4. Low Confidence Handling

If the model returns a low confidence score, the system:

- Requests additional clarification from the user
- Asks follow-up questions to improve decision accuracy
- Delays final decision-making until sufficient information is obtained

5. Graceful Error Handling

If the system is unable to generate a reliable output after retries:

- A clear message is displayed to the user
- The user is advised to seek medical attention directly
- Basic guidance (e.g. visit nearest clinic or hospital) is provided

6. Emergency Escalation Path

In high-risk scenarios or system uncertainty:

- The system prioritizes safety by recommending immediate emergency action
- Users are directed to emergency services without relying solely on AI output

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5. User Stories & Use Cases

- **User Stories:**
  - **As a patient**, I want to describe my symptoms in my own words so that I can receive an accurate assessment without medical knowledge.
  - **As a patient**, I want the system to ask follow-up questions when my input is unclear so that the diagnosis becomes more accurate.
  - **As a patient**, I want to know whether my condition is emergency, urgent, or non-urgent so that I can take the appropriate action quickly.
  - **As a patient**, I want to be guided to the right doctor and available time slot so that I can book an appointment efficiently.
  - **As a patient**, I want a clear summary of my condition and appointment so that I understand what to do next.

- **Use Cases (Main Interactions):**

  1. Symptom Input & Understanding Flow

  User enters symptoms via text → System processes input → extracts symptoms, duration, severity → Converts into structured medical data for further analysis.

  2. AI Triage & Classification Flow

  Structured data is passed to the AI Triage Agent → System evaluates urgency using AI + rule-based logic → Outputs:
  - Emergency / Urgent / Non-Urgent
  - Confidence score
  - Risk explanation

  3. Adaptive Follow-Up Flow

  If input is incomplete → System triggers Missing Info Agent → User answers follow-up questions → System updates state and refines triage result.

  4. Decision & Action Routing Flow

  Based on triage result:
  - Emergency → immediate alert + emergency guidance
  - Urgent → fast-track doctor recommendation

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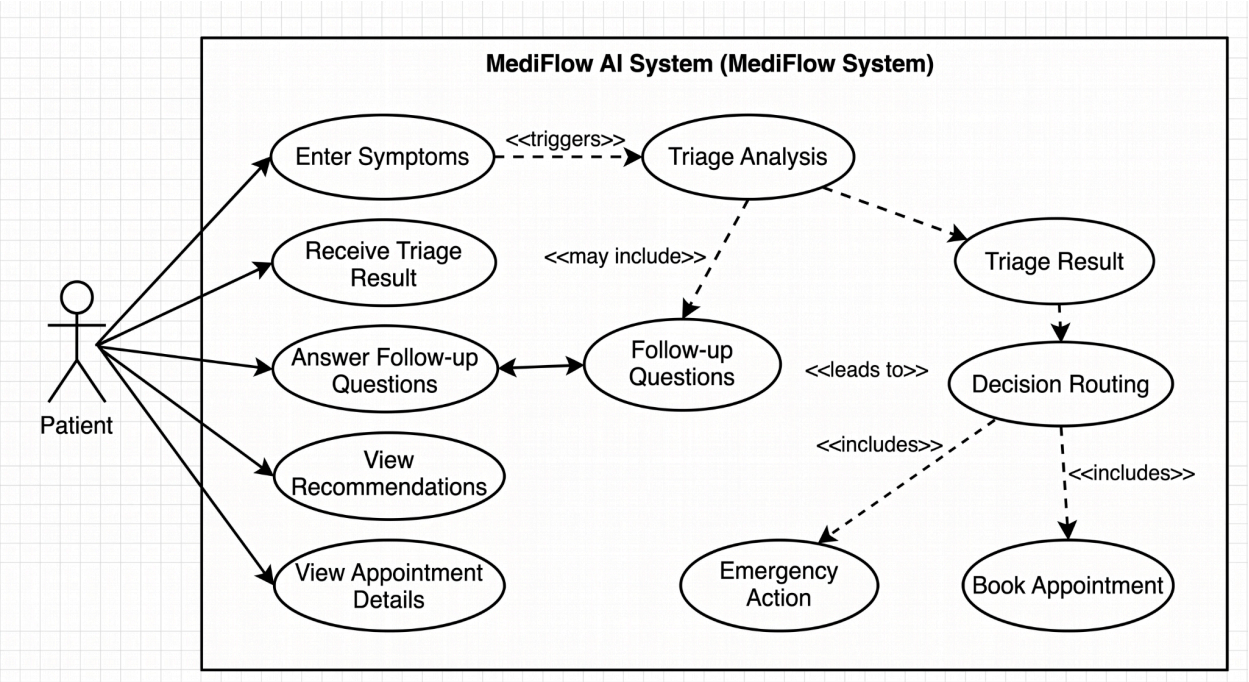
- Non-Urgent → normal appointment flow

5. Result & Summary Flow

System generates structured output:

- Patient summary
- Triage level
- Appointment details
- Next-step guidance

- **Use Case Diagram:**



6. Features Included (Scope Definition)

- AI-powered patient triage system that classifies user symptoms into urgency levels (e.g., Emergency, Urgent, and Non-Urgent).
- Conversational interface for users to input symptoms in natural language.
- Smart Input Understanding Agent that extracts key medical symptoms and intent from user descriptions.
- Automated recommendation of appropriate department (e.g., Emergency, General Clinic, Specialist).

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- Document upload support (e.g., medical reports, ID) for context-aware triage suggestions.
- Simple dashboard view for displaying triage results and suggested next actions.
- Clean and responsive UI optimized for both mobile and desktop usage.
- Real-time feedback generation based on AI analysis of user input.

7. Features Not Included (Scope Control)

- No real hospital backend integration (e.g., hospital appointment systems or EMR/EHR databases).
- No medical diagnosis or treatment decision-making (system only provides triage suggestions, not diagnosis).
- No real-time doctor consultation or video call integration in initial version.
- No multi-user hospital management system (e.g., staff scheduling, admin hospital operations).
- No production-grade security compliance (e.g., HIPAA-level certification).
- No integration with external government or insurance APIs in the prototype stage.
- No continuous patient monitoring or IoT medical device integration.

8. Assumptions & Constraints

- **LLM Cost Constraint:**
  - MediFlow assumes an average user session consists of symptom input, AI processing, triage classification, and recommendation output.
  - Estimated token usage per session: 1,000 – 1,800 tokens (including input + reasoning + response generation).
  - Based on this, the estimated cost per session is kept low by:
    - Limiting long conversational history storage (only last 3–5 interactions are considered).
    - Using summarised medical context instead of full chat logs for repeated reasoning.

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    - Implementing response caching for common symptom patterns (e.g., fever, headache, flu-like symptoms) to avoid re-calling the LLM for similar inputs.
  - Design decision for cost control:
    - The system prioritizes structured triage classification outputs (fixed format responses) instead of free-form long explanations, significantly reducing token usage per request.

- **Technical Constraints:**
  - The system is limited to frontend + AI workflow prototype only, without full hospital system integration.
  - No direct integration with real hospital databases, EMR systems, or government healthcare APIs.
  - AI output is restricted to:
    - Symptom classification
    - Department recommendation
    - Basic health guidance (non-diagnostic)
  - No support for advanced medical imaging analysis (e.g., X-ray, MRI interpretation).
  - Deployment is assumed on lightweight cloud services (e.g., serverless or basic API hosting), not enterprise-scale infrastructure.

- **Performance Constraints:**
  - To maintain fast response time, AI output is optimized for short structured responses instead of long medical explanations.
  - Complex symptom descriptions may require longer processing time, so a maximum response length limit is enforced.
  - High system load may result in delayed responses due to computation overhead.

- **User Input Constraints:**
  - The system is fully dependent on user-provided symptom descriptions and cannot operate without human input.

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- Users are expected to describe symptoms clearly; vague inputs may reduce accuracy of triage results.
- The system is not production-certified medical software, so all outputs are advisory and require human (medical professional) validation before any real decision-making.

9. Risks & Questions Throughout Development

1. **AI Accuracy & Reliability**
   - How do we ensure the AI provides consistent and safe triage recommendations, especially for critical cases?
   - What happens if the AI misclassifies symptoms (e.g., marking a critical case as low urgency)?
   - How do we validate that responses remain medically reasonable but non-diagnostic?
2. **Input Interpretation Risk**
   - What will happen if users provide vague, incomplete, or misleading symptom descriptions?
   - How does the system handle ambiguous inputs such as “I feel unwell” without sufficient details?
3. **System Consistency & Stability**
   - Will the AI generate consistent triage outputs for similar symptom inputs across different users?
   - How do we ensure the UI and triage results remain stable and readable across devices?
   - What happens if the system produces conflicting recommendations in different sessions?
4. **Safety & Ethical Concerns**
   - How do we prevent users from over-relying on the system instead of seeking real medical help?

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- What safeguards are in place to ensure critical cases are always escalated properly (e.g., advising immediate hospital visit)?
- How do we communicate clearly that the system is not a replacement for professional diagnosis?

5. **Performance & Scalability Risks**
    - What happens when there is high user traffic—will response time significantly increase?
    - Can the system maintain performance while keeping costs manageable at scale?
6. **Development & Integration Questions**
    - How will the system behave without real hospital backend integration?
    - What limitations arise from using a prototype-level architecture instead of production infrastructure?
    - How easily can the system be extended in the future to support real-world healthcare integrations?

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